

Ankit Shukla

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Senior Software Engineer | C/C++ | Linux Systems | Embedded Systems | Cloud Native

PROFESSIONAL SUMMARY

Senior Software Engineer with 14+ years building and shipping systems-level C/C++ software across Linux, Windows, QNX, and WinCE. Deep expertise in multithreaded and distributed systems, Linux internals, IPC, and socket programming, spanning storage, automotive, avionics, and graphics domains. Proven track record designing resource-management and licensing systems, delivering cloud-integrated services on AWS and Azure, and taking features from design through implementation, testing, and production on-call support. Experienced technical leader who has run a team of six, mentored engineers through code review and design discussions, and worked comfortably from device-driver and kernel-adjacent code up through cloud services.

PROFESSIONAL HIGHLIGHTS

- 14+ years in Systems Software Development
- Linux Systems Programming
- Multithreading & Concurrency
- Technical Leadership (led team of 6)
- Production Debugging & On-call Support
- Expert in Modern C++ (11/14/17)
- Embedded Linux & RTOS (QNX, WinCE, LynxOS)
- Distributed Systems
- Cross-platform Development
- Cloud-Integrated Infrastructure (AWS, Azure)

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Go, Java

Modern C++: STL, Templates, Smart Pointers, RAII, Move Semantics, Lambda Expressions, Design Patterns (Singleton, Factory, Adapter, Builder), SOLID

Systems Programming: Linux, POSIX, IPC, Socket Programming, TCP/IP, Memory Management, Multithreading, Synchronization, Performance Optimization

Embedded Systems: Embedded Linux, QNX, WinCE, LynxOS, ARM, x86, Kernel-adjacent Development, Device Drivers

Distributed Systems & Cloud: AWS, Azure, REST, gRPC, SOAP, JSON/XML, OpenSSL 3.0

Testing: GoogleTest, GoogleMock, VectorCAST, Ginkgo, TestRail

Build Tools: CMake, Make, Bazel, Apache Maven, MS Build, Jenkins

Debugging: GDB, Trace32, Lauterbach, Core Dump Analysis, Root Cause Analysis

Version Control & PM: Git, SVN, RTC, Bitbucket, Gerrit, Jira, IBM Rational DOORS

AI-assisted Development: Cursor, GitHub Copilot with Claude models

PROFESSIONAL EXPERIENCE

Cohesity Inc. Dec 2024 – Present

Software Engineer – Resource Manager (Heimdall) | Storage

- Built core components of Heimdall, an internal resource manager that provisions and tracks Linux/Windows VM fleets and disks used by backup services, in C++, Python, and Go.
- Owned features end-to-end: wrote design docs, drove requirement discussions with dependent teams, broke work into epics/stories in Jira, and implemented and code-reviewed the resulting changes.
- Wrote unit tests (GoogleTest/GoogleMock) and integrated automated test runs through Ginkgo, publishing results to TestRail for QA sign-off.
- Investigated and resolved release-blocking defects, including issues surfaced during on-call rotations for customer-impacting backup failures.

- Maintained the build/image scripts used to produce Heimdall deployment images across environments.

Veritas Technologies *Jan 2023 – Dec 2024*

Software Engineer – Shared Licensing Component (SLIC) | Storage

- Developed licensing/entitlement features in C++ for SLIC, the shared license-validation component used across Veritas's product line, built on OpenSSL and Poco.
- Designed and implemented on Windows, then ported and validated changes across Red Hat, OpenSUSE, AIX, and Solaris.
- Rebuilt the Poco and OpenSSL dependencies on each supported platform and delivered updated SLIC SDKs to downstream product teams to close known CVEs.
- Triaged vulnerabilities flagged by Black Duck in OpenSSL/Poco and assessed real-world impact on SLIC before prioritizing fixes.

Siemens Mobility *Mar 2019 – Jan 2023*

Senior Software Engineer / Technical Lead – Radio Block Center (RBC) | Rail / Mobility

- Led a team of 6 engineers on RBC, safety-critical software that receives train position reports over Balise devices and issues movement authority to trains.
- Delivered new features in C/C++/Python from requirements through implementation, static analysis, and unit testing with VectorCast.
- Ran code reviews and diagnosed field issues from logs, driving root-cause analysis and fixes for the team.
- Built two internal automation tools in Python (Tkinter) — one to automate environment/software setup, another to update code baselines and timestamps across unit-test suites — both adopted team-wide to cut manual setup time.
- Owned monthly billing releases and forecast updates for the engagement.

Tata Elxsi Ltd *Jun 2015 – Mar 2019*

Software Engineer – Automotive Infotainment & Instrument Cluster (QNX) | Automotive

- Applink (Panasonic Automotive, 3 yrs 2 mo): built C/C++ modules for a smartphone-to-infotainment connectivity system on QNX/Windows, covering voice command handling, touch UI input, and embedded navigation; authored design docs, analyzed core dumps, and fixed defects found in production logs.
- GM Instrument Cluster (Visteon, 6 mo): processed vehicle bus data and drove UI widgets showing tire pressure, fuel level, and speed/tachometer readings on QNX/Windows; delivered features from system design through GoogleTest/GoogleMock unit testing and static analysis.
- Used Singleton and related design patterns to keep shared hardware-access code consistent across both projects.

Infosys Ltd *Aug 2014 – Jun 2015*

Software Engineer – Graphics Driver Development (AMD, WinCE) | Graphics / Multimedia

- Ported the Linux OpenGL/LLVM call stack to WinCE and worked on the Radeon GPU driver's DRM subsystem to bring hardware-accelerated rendering to WinCE-based boards (Ontario, Kabini, Mullins).
- Diagnosed and fixed driver defects affecting video and gaming performance, and built OpenGL test applications to validate rendering correctness.

Rockwell Collins India Pvt. Ltd *Aug 2011 – Aug 2014*

Software Engineer – Onboard Maintenance System (RTOS LynxOS) | Avionics

- Developed features in C/C++ for OMS, an avionics subsystem (Diagnostic Report and Display Manager applications) that communicates with aircraft LRUs over ARINC/Ethernet and XML-RPC to surface maintenance data to flight crews.
- Worked on a 20-person program using Trace32/Lauterbach for hardware-level debugging on LynxOS targets, with requirements tracked in IBM Rational DOORS.
- Wrote and executed functional and unit tests and fixed defects found during integration testing.

EDUCATION

B.Tech — UPTU, 2010

